IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 5: Thresholding Techniques

Fill in the table:

Global Thresholding – Used When: The image has a uniform illumination | Advantage: Simple and fast

Adaptive Thresholding – Used When: The image has a uneven illumination | Advantage: Works well with local variations in brightness

Otsu Method – Used When: The image has two distinct intensity regions (bimodal histogram) | Advantage: Automatically selects the optimal threshold

Why does global threshold fail under uneven illumination?

Global thresholding uses **one fixed threshold value for the entire image** to separate the foreground from the background. This method works well when the image has **uniform or consistent illumination**, because the foreground and background have relatively stable brightness levels. However, when the image has uneven illumination, different parts of the same image may have different brightness levels.

For example, one part of an object may be brightly illuminated while another part may be in a shadow. If a single threshold is applied to the whole image, the bright region may be correctly identified as the foreground, but the darker region of the same object may be incorrectly classified as the background. Similarly, a dark background region may sometimes be classified as part of the object. This results in **incorrect segmentation and loss of important image information**.

The main problem is that a **single threshold cannot adapt to local changes in brightness**. Uneven lighting can occur because of shadows, reflections, changes in light intensity, or the position of the light source. Therefore, pixels with similar object information may have very different intensity values in different areas of the image.